A shorter complete consistency equation

Fifteen strictly positive integer unknowns; all fixed constants defined on this page.

The unknowns are

$$J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma \in \mathbb{Z}_{>0}.$$

Use the following polynomial abbreviations (no extra unknowns):

$$\begin{gathered}
q = \beta J + 1, \quad C = q - F - \alpha, \quad X = wq, \\
Y = sq^3, \quad E = XY, \quad k = \eta + \zeta, \quad c = kY + \eta, \\
a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \\
D = X + ac + \gamma H, \\
R = ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J, \\
V = c(Tf - 1) - Rf^2.
\end{gathered}$$

The equation is

$$\boxed{N_0 N_1 N_a N_k N_t N_s - 1 = 0,}$$

where

$$\begin{aligned}
N_0 &= \tau^2 - (XY^2)(XY^2 + 1)k^2, \\
N_1 &= D^2 - \Delta c^2, \\
N_a &= \Delta(f^2 - 1)(V^2 - y^2) + y^2, \\
N_k &= k - hE - R, \\
N_t &= (K + w)C + (q - F) - z(q - 1), \\
N_s &= f^2 - \Delta(ic^2)^2.
\end{aligned}$$

Fixed coefficients. All sums below have the displayed fixed finite limits. Division and floor are used only to define constants; the equation above is polynomial. Put

$$\begin{gathered}
\rho = 2^{89}, \quad B = \rho^{4831}, \quad \beta = B - 1, \\
\lambda = 2^{29}, \quad \theta = 2^{42}, \quad Q = 2^{40},
\end{gathered}$$

$$\begin{gathered}
G_m(x) = \sum\nolimits_{j=0}^{m-1} x^j, \\
\mu = 510 + 2^{17} + 2^{26} + 2^{28} + \lambda(2^{44} - 1).
\end{gathered}$$

Let

$$\begin{aligned}
\mathcal{S} = {} & \{0, \dots, 456\} \cup \{495, \dots, 504\} \cup \{507, 508\} \\
& \cup \{517, \dots, 537\} \cup \{539, \dots, 549\} \\
& \cup \{567, \dots, 583\} \cup \{1203, 1807\}.
\end{aligned}$$

Then

$$\begin{aligned}
M_C &= B - 1 - \sum\nolimits_{e \in \mathcal{S} \setminus \{1\}} \rho^e - 2\rho^{507}, \\
M_F &= B - 1 + 4 + \mu(\rho^{2409} + \rho^{3015}) \\
&\quad + \rho^{528} G_{10}(\rho) + \rho^{585} G_{11}(\rho) + \rho^{1203} + \rho^{1807} \\
&\quad + (2^{87} - 1)\rho^{4823} G_4(\rho) + (2^{38} - 1)\rho^{4827}, \\
K &= \rho^{33} + \rho^{68} + \rho^{664} + \rho^{1268} + \rho^{4877} \\
&\quad + (\rho^{2409} + \rho^{3015})\Phi(1/\rho),
\end{aligned}$$

where the rational intermediate expression is

$$\begin{aligned}
\Phi(x) = {} & \sum\nolimits_{i=0}^{456} c_i x^{r_i} \\
& + \lambda\big[(x^{539} + 2^{21}x^{517})G_{11}(2x) \\
& \qquad + (2^{11}x^{528} + 2^{32}x^{495})G_{10}(2x) \\
& \qquad + \theta(x^{539} + x^{1203} + x^{1807})\big] \\
& + \big[(2^9 + \lambda(2 + 2^{33}))x^{567} \\
& \qquad + (2^{18} + \lambda(2^{22} + 2^{33}))x^{575}\big] G_8(2x) \\
& + (2^{27} + \lambda(2^{11} + 2^{32}))x^{583}.
\end{aligned}$$

The resulting β, K, M_C, M_F are fixed positive integers.

The constants r_i, c_i. First define the following 457 fixed quadruples $t_i = (w_i, n_i, e_i, s_i)$. For fixed indices $j = 2q + r$, $0 \le q < 215$, $r \in \{0, 1\}$, put

$$\begin{gathered}
D_b(A, k) = \left\lfloor A/b^k \right\rfloor \bmod b, \\
\mathcal{N} = \mathcal{A} \bmod 2^{426}, \quad \mathcal{T} = \left\lfloor \mathcal{A}/2^{426} \right\rfloor \bmod 2^{1675}, \\
\mathcal{M} = \left\lfloor \mathcal{A}/2^{2101} \right\rfloor \bmod 2^{206}, \quad \mathcal{E} = \left\lfloor \mathcal{A}/2^{2307} \right\rfloor, \\
\nu_j = D_2(\mathcal{N}, j), \quad \sigma_j = \sum\nolimits_{k=0}^{j-1} \nu_k, \\
m_q = D_2(\mathcal{M}, q), \quad \tau_q = \sum\nolimits_{k=0}^{q-1} m_k, \\
e_q = D_{12}(\mathcal{E}, \tau_q), \\
F_q = 14(1 - m_q) + m_q\big(e_q + \lfloor e_q/9 \rfloor + \lfloor e_q/10 \rfloor + 2\lfloor e_q/11 \rfloor\big), \\
u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j) D_{216}(\mathcal{T}, j - \sigma_j), \\
v_{2q+r} = D_2(F_q, 2r), \quad d_{2q+r} = D_2(F_q, 2r + 1).
\end{gathered}$$

Empty sums are zero; $a \bmod b = a - b\lfloor a/b \rfloor$. Set

$$t_{2+j} = \big((1 - d_j)(512 + u_j),\ 512 + j,\ d_j(256 + u_j),\ v_j\big).$$

The remaining quadruples are

index	range	t_i
0		$(2, 3, 3, 1)$
1		$(5, 0, 0, 0)$
$432 + j$	$0 \le j < 2$	$(0, 942 + j, 0, 942 + j)$
$434 + j$	$0 \le j < 2$	$(0, j, 0, j)$
$436 + j$	$0 \le j < 4$	$(0, (j \bmod 2) + 942\lfloor j/2 \rfloor, 0, 6)$
440		$(2, 2, 2, 0)$
441		$(3, 4, 4, 698)$
442		$(4, 4, 4, 0)$
$443 + j$	$0 \le j < 3$	$(1, 6, 1, 2 + j)$
$446 + 3j + k$	$0 \le j, k < 3$	$(j^2, 5, k, 5)$
455		$(256, 0, 0, 512)$
456		$(0, 0, 512, 512)$

Define, for these fixed indices,

$$\begin{aligned}
\kappa_i &= 2^{30}w_i + 2^{20}n_i + 2^{10}e_i + s_i, \\
\ell_i &= 2w_i + 2^{11}n_i + 2^{22}e_i + 2^{32}s_i, \\
r_0 &= 0, \quad r_1 = 1, \\
r_i &= 2 + \sum\nolimits_{j=2}^{456} \left\lfloor \frac{Q + \kappa_i - \kappa_j - 1}{Q} \right\rfloor \quad (2 \le i \le 456), \\
\delta_{ij} &= \left\lfloor \frac{Q - (i - j)^2}{Q} \right\rfloor, \\
O &= 1 + 255\,2^9 + 255\,2^{18} + 2^{27} + \lambda, \\
c_i &= O + \lambda\ell_i - 255\,2^9\delta_{i,455} - 255\,2^{18}\delta_{i,456} \\
&\qquad - 2^{27}(\delta_{i,455} + \delta_{i,456}).
\end{aligned}$$

All subscripts in these definitions are bound constant indices, not additional Diophantine unknowns.

The integer $\mathcal{A}$. Concatenate the following lines in order:

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669131531036130207918705424063251935678361402127158315939485900
024862121683918620497399261763604270047463665495184619310736400
634168923026995163985631137298551907531755581759869092324990250
245354881915218824145172166779625738892477812354929944560201810
714979081103521527996053187219897902401443655603596614605431430
401238213136877764687031187676120608383410762420136492629591520
948105430088283992977776624494899927891143404043188233946502060
633863435783318094517854835611729680930405601146271914846583700
522719662693677541580402696904889134177240630983710085469908080
128290547887619151486405782836404475361648121672814747436072840
456628770525259483183122602813360333006111441424362232664171360
556843059640886074087524846775179792895529277051522044452874950
5057997261564539069824012690
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The state-renamed native/P69 argument gives unsolvability iff Con(ZFC). Native calibration and its internal ZFC proof remain open in Lean. This page specifies the equation.